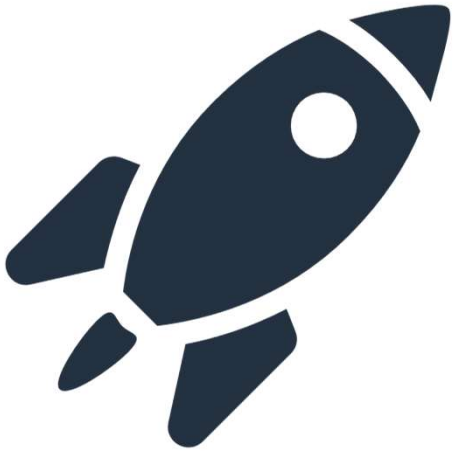
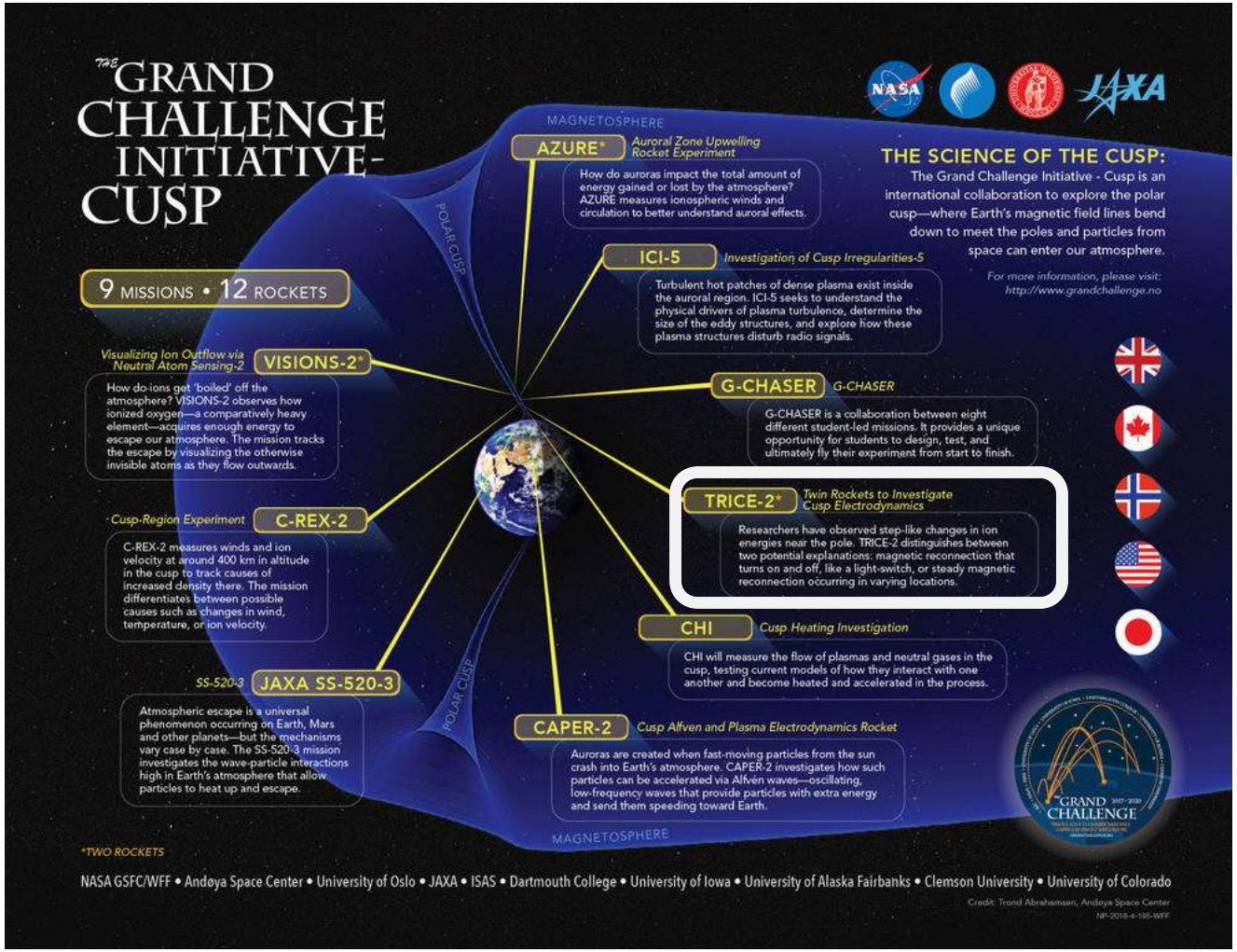
SPACE SCIENCE INSTRUMENTATION

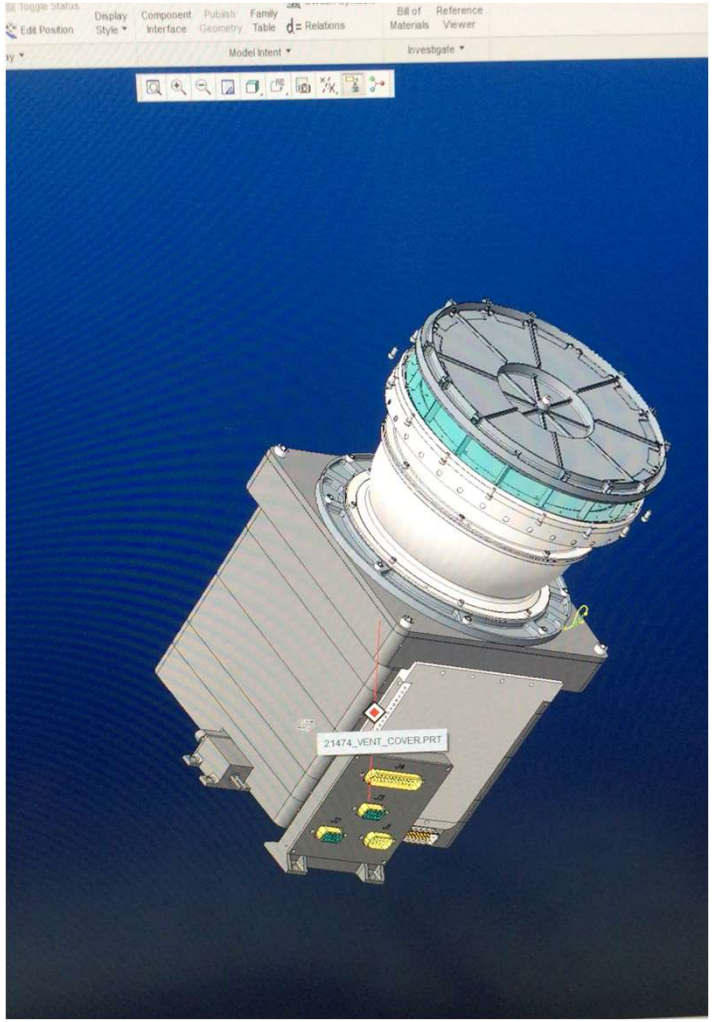
APIS II INSTRUMENTS

Twin Rockets to Investigate Cusp Electrodynamics-2 (TRICE-2)

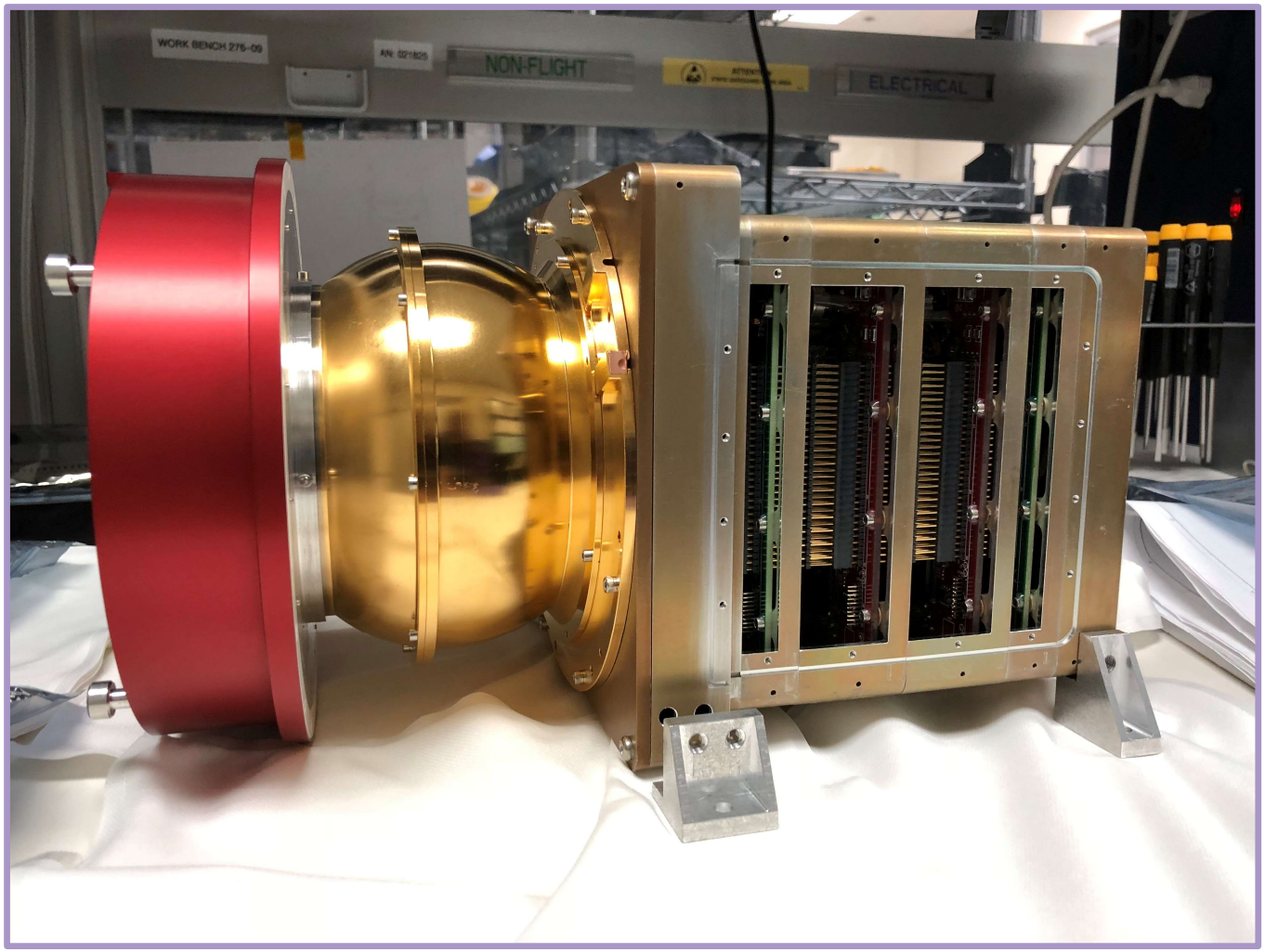




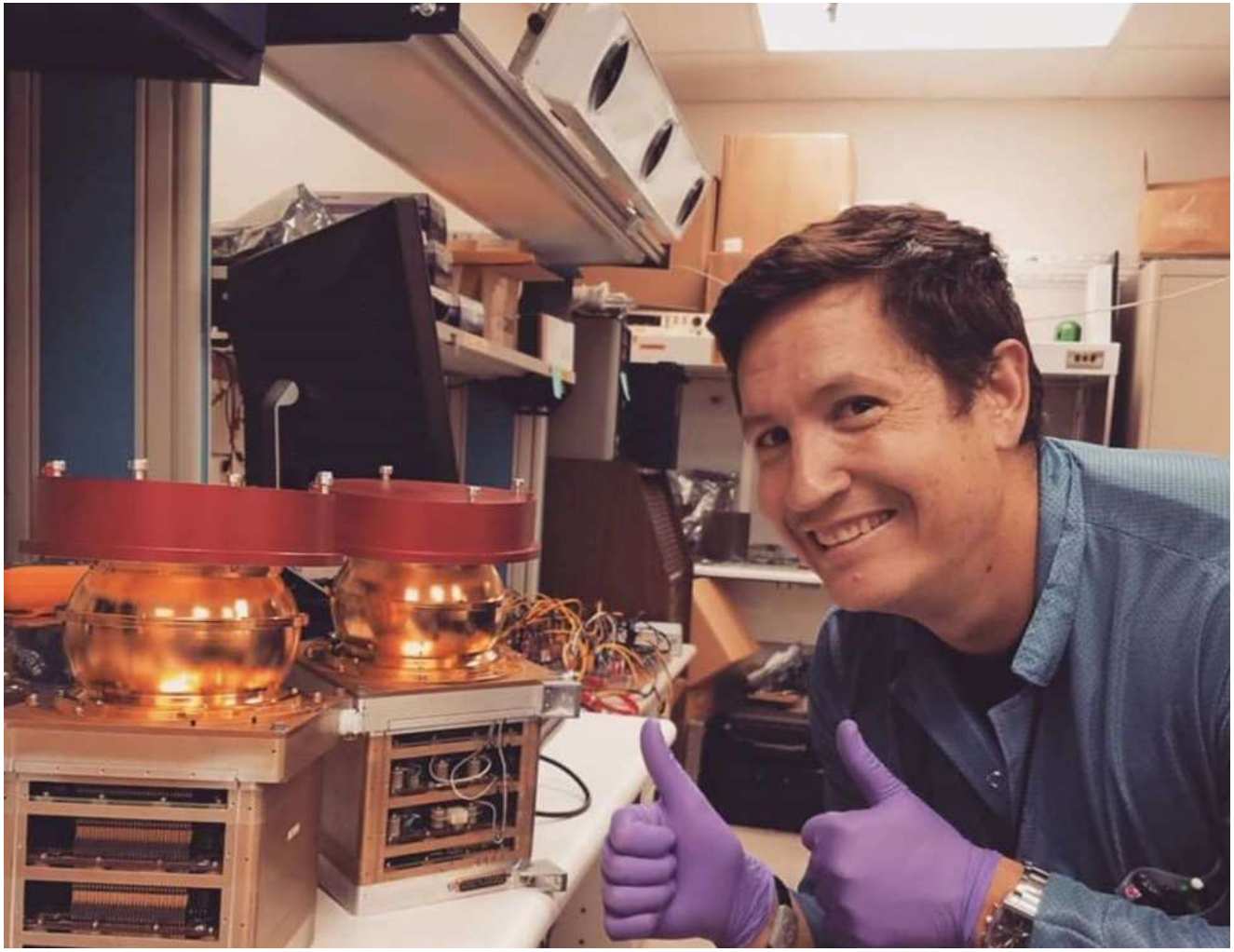
- Launching of two nearly identical instrumented payloads
- Measure cusp signatures of reconnection occurring at the magnetopause



- Lead Mechanical Engineer
- Critical Design Requirements
- Deployment Mechanism
- Shock Dampening



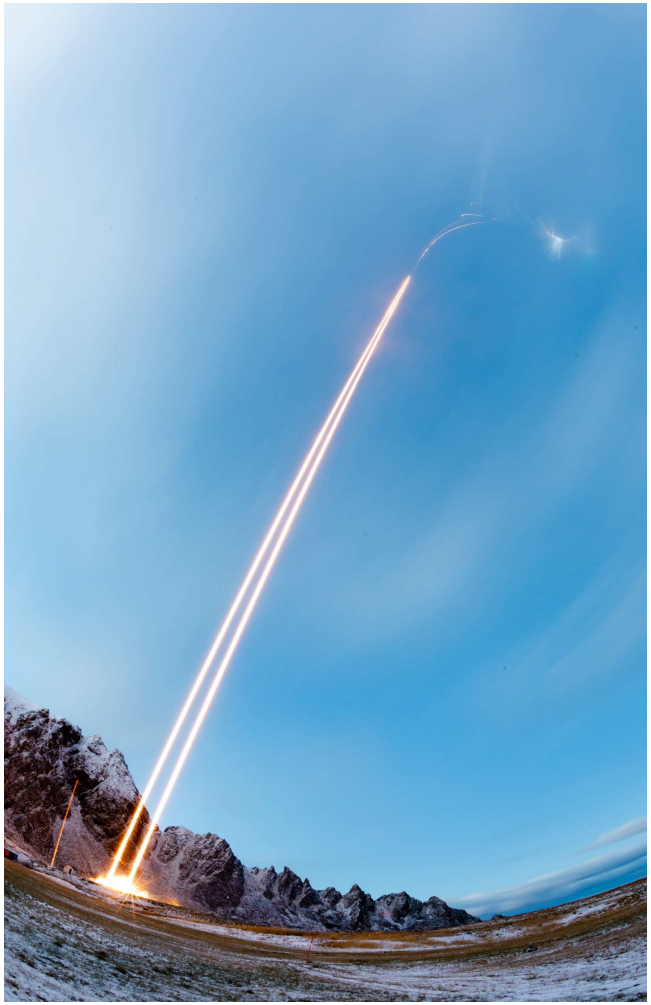
- Hot Plasma Composition Analyzer (HPCA)
- Collimator Grid
- High voltage boards
- Deployment mechanism



Lead Electrical Engineer with APIS II instruments. Image courtesy of SwRI Facebook Page

Challenges

- High voltages
- MCP sensitivity
- Board accessibility
- Loads for deployment
- Mechanical assembly procedure



TRICE-2 APIS II

SUCCESSFULLY LAUNCHED & DEPLOYED

DECEMBER 8TH, 2018